

Reading

PREPARING TO READ

A BUILD VOCABULARY The words in **blue** are used in the reading passage. Read the text below. Then write the correct form of each word next to its definition.

Around the world, the pigeon has become **synonymous** with city life. Derived from the rock dove—a similar bird that builds its nest high up on rocky cliffs—it is no wonder pigeons have adapted so well to urban life. Tall buildings with high ledges and small open spaces make an ideal **habitat** for these resourceful birds. The food humans leave behind provides them with a rich and constantly replenishing source of nourishment. And the relative lack of **predators**—like hawks and eagles—means the species is able to thrive and reproduce freely.

Experts who track pigeon populations estimate that there are up to 400 million pigeons living in cities worldwide. Unfortunately, these birds are usually **perceived** as troublesome pests because of the waste they leave behind and the diseases they spread. In order to **minimize** pigeon population growth, some cities like Venice, Bangkok, and Singapore have imposed fines for feeding pigeons. Obviously, this alone won't be enough to completely **eradicate** the problem. In all likelihood, pigeons are here to stay. The best thing we humans could probably do would be to **embrace** them, and adapt to living with them the same way they have adapted to living with us.

1. _____ (v) to reduce the extent of something
2. _____ (v) to accept something enthusiastically
3. _____ (v) to put an end to something
4. _____ (v) to think of something a certain way
5. _____ (adj) closely associated with or connected to something
6. _____ (n) an animal that hunts other animals for food
7. _____ (n) the natural home of a plant or animal species

A pigeon on a ledge
in New York City



B BUILD VOCABULARY Complete the sentences below with the correct form of the words and phrases in **blue**. Use a dictionary to help you.

boundary (n)
hypothesis (n)

consistently (adv)
reliance (n)

constraint (n)
reversal (n)

counterpart (n)
stem from (v)

1. Her idea about the changing migration patterns of some of the birds in this city has yet to be proven. For now, it's just a(n) _____.
2. Together, the river and the mountain range form a clear and natural _____ that separates the two countries from each other.
3. Many of the locals feel that most of the city's problems _____ corrupt government officials.
4. In a stunning _____, the city council announced its decision not to go ahead with the renewal project.
5. Some people regard the pigeon as the flying _____ of the rat. They both thrive in cities and live on trash.
6. To improve traffic in the city, we need to reduce our _____ on cars by developing an efficient and comfortable public transportation system.
7. Kyoto is _____ ranked as one of the best cities to visit in Japan.
8. To grow, the city needs to overcome some natural _____, like limited space and fresh water.

C USE VOCABULARY Discuss these questions with a partner.

1. What are some animals that are commonly found in cities? What makes cities such great **habitats** for these animals?
2. How are the animals that live in your town or city **perceived**? Are they well liked or thought of as pests?
3. How can the problem of pests in a city be **minimized**? Is it possible to **eradicate** pests completely? Why or why not?
4. Are there any wild **predators** that live in your town or city? How much of a threat are they to the people who live there?

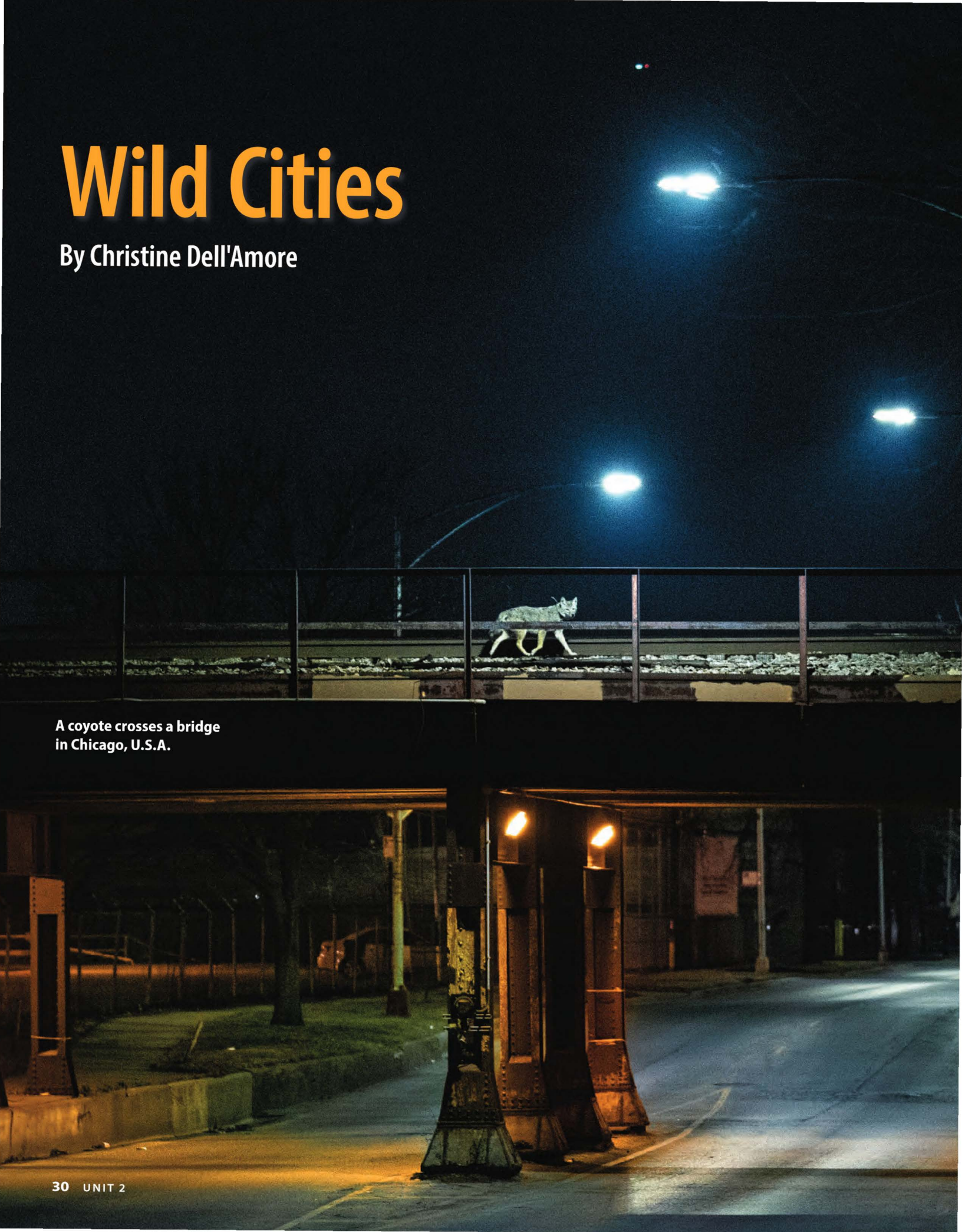
D PREDICT Look at the photos in the reading passage and read the first sentence of each paragraph. Then answer the questions. Check your answers as you read the passage.

Critical Thinking

1. What urban challenge does the reading passage discuss?
2. How do you think this issue affects people living in cities?
3. What solution(s) to the challenge do you think the reading passage offers?

Wild Cities

By Christine Dell'Amore

A full-page photograph of a coyote walking across a bridge at night. The coyote is silhouetted against the dark background of the bridge deck. The bridge has a metal railing. In the background, there are streetlights and the structure of the bridge's supports. The scene is illuminated by the cool blue light of the streetlights and the warmer yellow light of the bridge's support lights.

A coyote crosses a bridge
in Chicago, U.S.A.

Coyotes, bears, raccoons, and other animals are adapting to urban life, resulting in increased contact with humans.

- A**  At first glance, it's a scene that plays out daily in cities across America. A U.S. Postal Service carrier steps out of his mail truck and strides across the street, letters in hand. That much is unremarkable. But this postman either doesn't notice or doesn't seem to care that a hefty American black bear, likely a young male, is sitting a few yards away, vigorously scratching his shedding winter coat.
- B** Immediately to the left, Interstate 240 roars behind a chain-link fence, apparently just white noise to the bear, which eventually wanders down the sidewalk deeper into this neighborhood barely a half mile from downtown Asheville, North Carolina.
- C** Along the highway, a team of researchers with the North Carolina Urban/Suburban Bear Study is captivated by another discovery: a deep hollow inside a gnarled silver maple tree. Bear N209, a radio-collared female that's among more than a hundred bears being tracked in a study, hibernated¹ there over the winter, despite the constant rush of vehicles mere feet away.
- D** "These bears still surprise me," Colleen Olfenbuttel, the state's black bear biologist, shouts over the din of traffic. She holds a ladder steady as a colleague scrambles inside the tree and measures the den. It's the biggest tree den Olfenbuttel has seen in her 23 years of studying black bears. "They're so much more adaptable than we give them credit for."
- E** Indeed, it's hard to imagine that black bears would take so well to living in Asheville. In this city of about 95,000, nestled in the Blue Ridge Mountains, bears shuffle down residential streets in broad daylight and climb onto people's decks and front porches. Some Asheville residents have **embraced** their furry neighbors, and nearly every person you talk with has a video of their most recent bear encounter.
- F** The advent of the city bear in Asheville and elsewhere **stems from** a combination of trends, including changes in land use and the tempting buffets² available when living near people. These factors have boosted North America's black bear population to nearly 800,000. At the same time, sprawling cities and suburbs have swallowed up large areas of bear **habitat**, leaving the animals little choice but to adapt to living with human neighbors.
- G** It's a phenomenon happening in urban areas around the world, and it's not unique to black bears. Many mammals that eat a wide variety of foods are moving in and changing their behaviors as they learn urban survival skills.
- H** Unfortunately, humans and bears don't always live in harmony—even in open-minded Asheville, where bears have killed pets and injured at least one person in recent years. In 2020, a mother bear defending her cubs attacked Valerie Patenotte's dog, which later died. "We understand everyone has to coexist," says Patenotte as we stand on her back deck overlooking the distant mountains. "We just want more space from bears."

¹ If an animal **hibernates**, it sleeps through winter to make it through the cold season.

² A **buffet** is a large spread of food. People pay a predetermined amount to eat as much of the food as they want.

A dumpster-diving bear rummages for food in South Lake Tahoe.



I While black bears have reclaimed about half their former range and now live in some 40 states, coyotes have taken the U.S. by storm in recent decades. They now can be found in every state except Hawaii, and in most major cities. The metropolis most **synonymous** with the urban coyote is Chicago, home to as many as 4,000 of the animals.

J Stan Gehrt, a wildlife ecologist with Ohio State University and the Max McGraw Wildlife Foundation, began studying Chicago's coyotes in 2000, not long after the animals started showing up there. Back then, Gehrt thought his project would last a year. More than two decades later, he's still at it. "We **consistently** underestimate this animal and its ability to adjust and adapt," Gehrt says. "They push the **boundaries** of what we **perceive** to be **constraints**."

K At the beginning of Gehrt's research, he thought coyotes would be restricted to parks and green spaces, but he was wrong. "Now we have coyotes everywhere—every neighborhood, every suburban city, and downtown."

L Indeed, coyotes have succeeded despite our best efforts to **eradicate** them. At least 400,000 are killed each year, about 80,000 by a federal **predator** control. Vehicle strikes are the main cause of death for Chicago's coyotes, but the animals have learned to avoid cars and can even read stoplights.

M Adding to their adaptability is their flexible diet. Coyotes will eat just about anything, from shoe leather to fruit (they can climb fruit trees). According to Gehrt, "Variability³ is the primary pattern."

N Christopher Schell, an urban ecologist at the University of California, Berkeley, and Julie Young, a wildlife biologist with the U.S. Department of Agriculture, are studying how various diets given to captive coyotes could change the animals' behavior. Their **hypothesis** is that the coyotes eating human food will become bolder around people, which is supported by some anecdotal evidence.⁴ Schell and Young theorize that a coyote that eats processed cereal,

³ Variability refers to a lack of consistency or predictable patterns.

⁴ Anecdotal evidence is evidence that is based on just a few examples or anecdotes (things people say). It is not based on science or data.

for instance, will be hungrier and look for food much more frequently than one that eats a nutrition-dense rabbit for breakfast. Though Gehrt has not noticed such a connection in Chicago coyotes, he notes that a **reliance** on human food does bring coyotes closer to restaurants and homes, which in turn leads to more contact and conflict with people and their pets.

- o When zoologist Sarah Benson-Amram first started looking into raccoon behavior and cognition⁵ about a decade ago, she figured such a common species would have been studied thoroughly. After all, the bushy-tailed omnivores⁶ are pop culture icons, jokingly dubbed trash pandas. Instead, Benson-Amram was shocked to find almost nothing in the scientific literature. A few researchers in the early 1900s had tried to study the clever animals, but gave up when their subjects kept breaking out of their cages.
- P So far, she says, her research has confirmed the raccoon's crafty reputation. In an experiment called **reversal** learning, she presented raccoons with a box equipped with a button or foot pedal that, when pressed, released food. After the animals figured out how to get the food, the researchers would switch the buttons and pedals, forcing them to tweak their strategy. Most of the raccoons were able to solve the problem on the first night. To put that into perspective, only one of six coyotes engaged with the box—and not until the 44th night of testing.

⁵ Cognition refers to how people or animals think and process information in their brains.

⁶ Omnivores are animals that eat a variety of plant- and animal-based foods.

In San Francisco, U.S.A., Misto the dog wears a spiked vest designed to deter hungry coyotes.



- Q** According to Benson-Amram, now at the University of British Columbia in Vancouver, raccoons have a different survival strategy from coyotes: they're successful by exploiting humans, not avoiding them.
- R** Like coyotes and bears, raccoons are expanding throughout North American cities. In Washington, D.C., wildlife researchers Kate Ritzel and Travis Gallo wanted to find out whether raccoons living in the city are bolder and more willing to take risks than those in rural areas. They measured this by observing a raccoon's readiness to investigate an unfamiliar object—in this case, bait⁷ buried inside a square of wooden stakes.
- S** The researchers installed more than a hundred automatic cameras throughout the city and rural areas of neighboring Virginia. On a muggy September morning at Fort Totten, Gallo placed the smelly bait—"dead animals in a jar," he called it—while Ritzel strapped a camera to a nearby tree. She would check the videos every two weeks to see which animals had passed through. Her favorite video? A feisty⁸ raccoon chasing off a fox.

⁷ Bait is something (typically food) used to lure animals into a trap.

⁸ A feisty person or animal is one that is small, fiery, and full of energy.



A mother raccoon emerges from a six-inch space between buildings where she is raising her young.

T Months later, Ritzel's data indicated that urban raccoons are more exploratory than their country cousins, taking more time to investigate unusual objects. City raccoons are also more social, traveling in pairs more often than their rural, more territorial **counterparts**—suggesting that urban raccoons are adapting their behavior to city life.

U Until recently, urban wildlife was mostly ignored in scientific research. This is partly because such species are considered pests unworthy of our attention—or not wildlife at all. However, according to Seth Magle—director of the Urban Wildlife Institute at Chicago's Lincoln Park Zoo—we live on a planet that's rapidly urbanizing. It would therefore be unwise to ignore animals that move into urban landscapes. Magle adds that while much of urban ecology focuses on how to **minimize** conflicts with these animals, we forget that many of our encounters with wildlife are delightful. For Magle, “Another part of coexisting with animals has to do with celebrating these moments.”

Adapted from “Why Cities Are Going Wild,” by Christine Dell’Amore: National Geographic Magazine, July 2022

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UNDERSTANDING THE READING

A UNDERSTAND MAIN IDEAS Choose the correct answers to the questions below.

- Which of the following is a reason more bears are moving into cities?
 - There are fewer predators in the city.
 - Cities are encroaching into bear territory.
- Why are urban coyotes a problem?
 - They regularly encounter people as they look for food.
 - They prefer human food to food available in the wild.
- How are urban raccoons different from urban coyotes?
 - Raccoons survive by limiting close contact with humans.
 - Raccoons survive by staying in close contact with humans.

B UNDERSTAND DETAILS Read the sentences. Choose **T** for true, **F** for false, or **NG** for not given.

- | | | | |
|--|----------|----------|-----------|
| 1. Olfenbuttel has studied bears for more than two decades. | T | F | NG |
| 2. The bear that attacked Valerie Patenotte’s dog died. | T | F | NG |
| 3. The coyote population in Chicago is around 40,000. | T | F | NG |
| 4. More coyotes are killed by vehicles than by federal predator control. | T | F | NG |
| 5. Coyotes are bigger than most dogs, but smaller than wolves. | T | F | NG |
| 6. Raccoons are better than coyotes at solving problems. | T | F | NG |
| 7. Rural raccoons are more social and exploratory than urban raccoons. | T | F | NG |

Critical Thinking

C SUMMARIZE Complete the chart with information from the passage.

Animal	Traits	Problems this might cause
bears		
coyotes		
raccoons		